

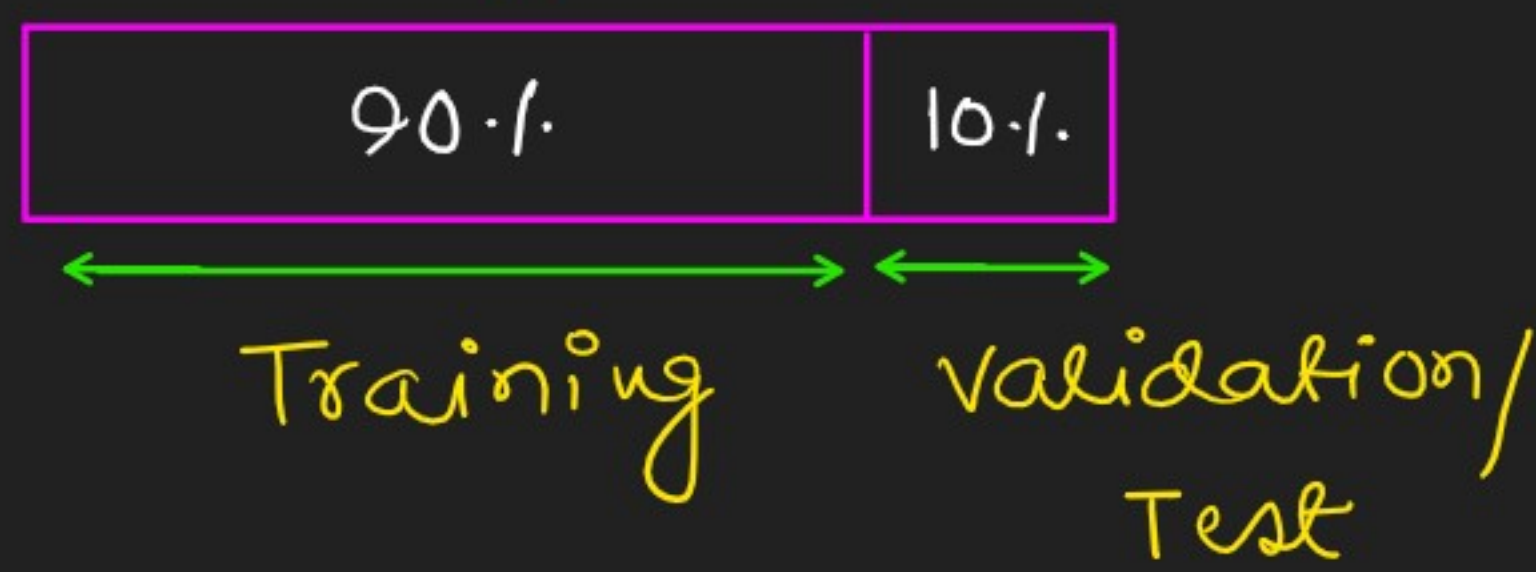
Lecture - 27: LLM Training and

Validation loss

Dataset used $\longrightarrow$ The verdict

Divide the dataset into training and validation.

```
train_ratio = 0.90  
split_idx = int(train_ratio * len(text_data))  
train_data = text_data[:split_idx]  
val_data = text_data[split_idx:]
```

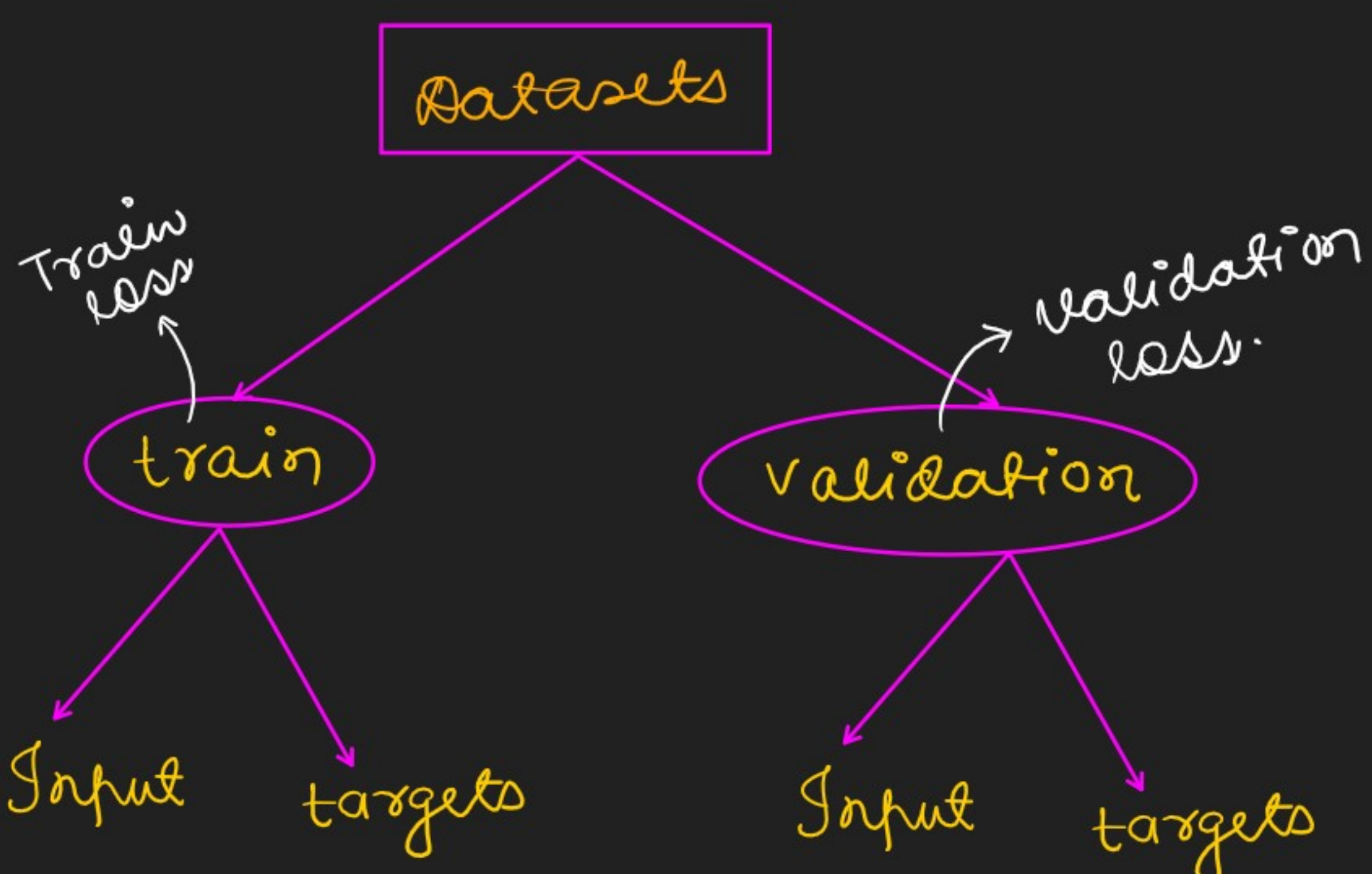


- use DataLoaders to chunk training and validation data into and output datasets.

Input: $x = \begin{bmatrix} \text{I had always thought} \end{bmatrix}$

Output: $y = \begin{bmatrix} \text{had always thought Jack} \end{bmatrix}$
 $\downarrow$
Here some data are loss.

context_side = 4 = stride = 4

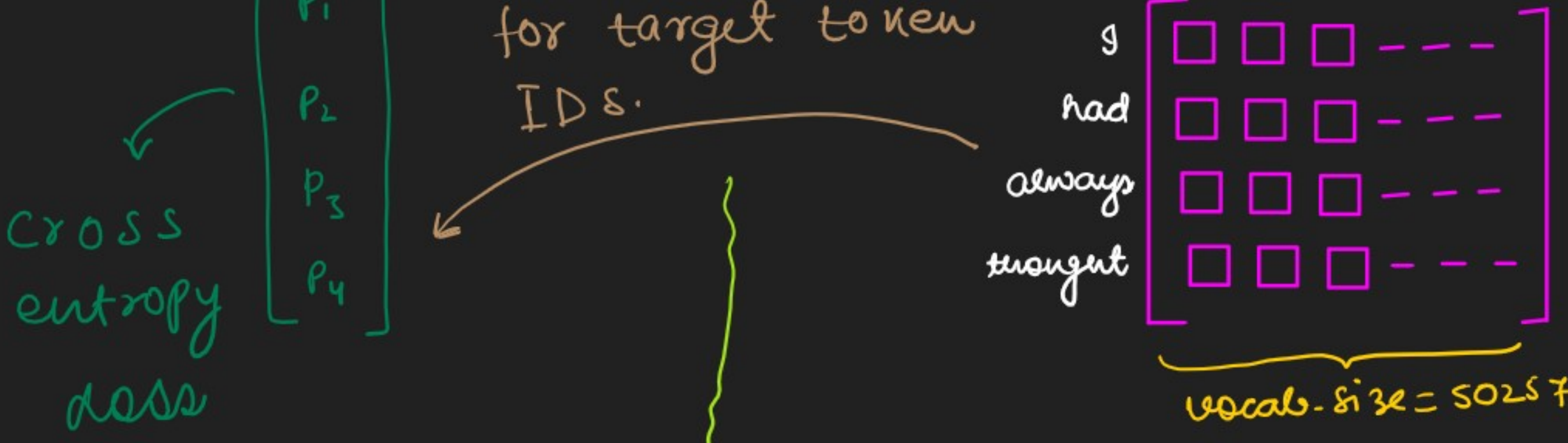
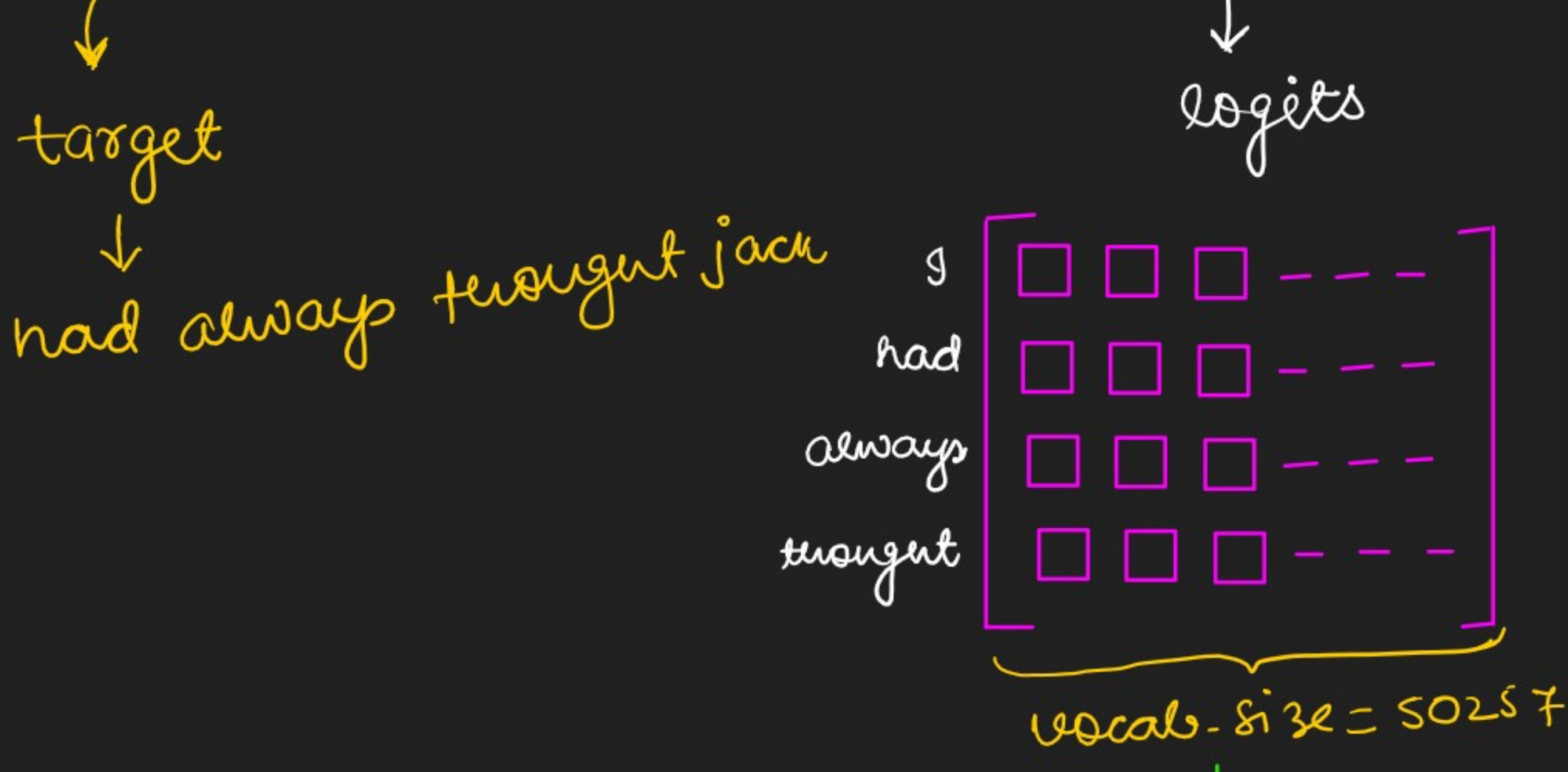


Input: $x = \begin{bmatrix} \text{I had always thought} \end{bmatrix}$

Output: $y = \begin{bmatrix} \text{had always thought Jack} \end{bmatrix}$

let's look at the first input

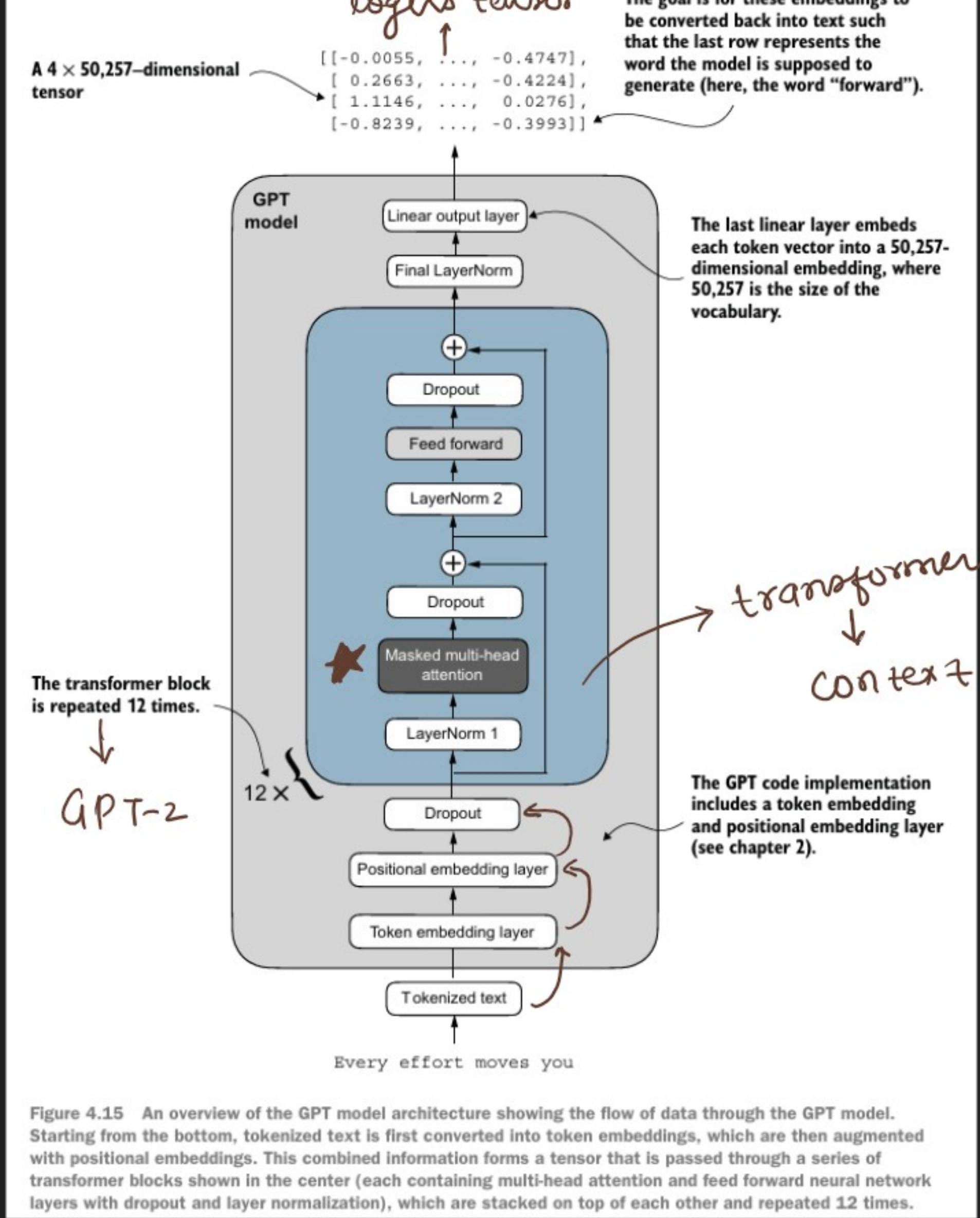
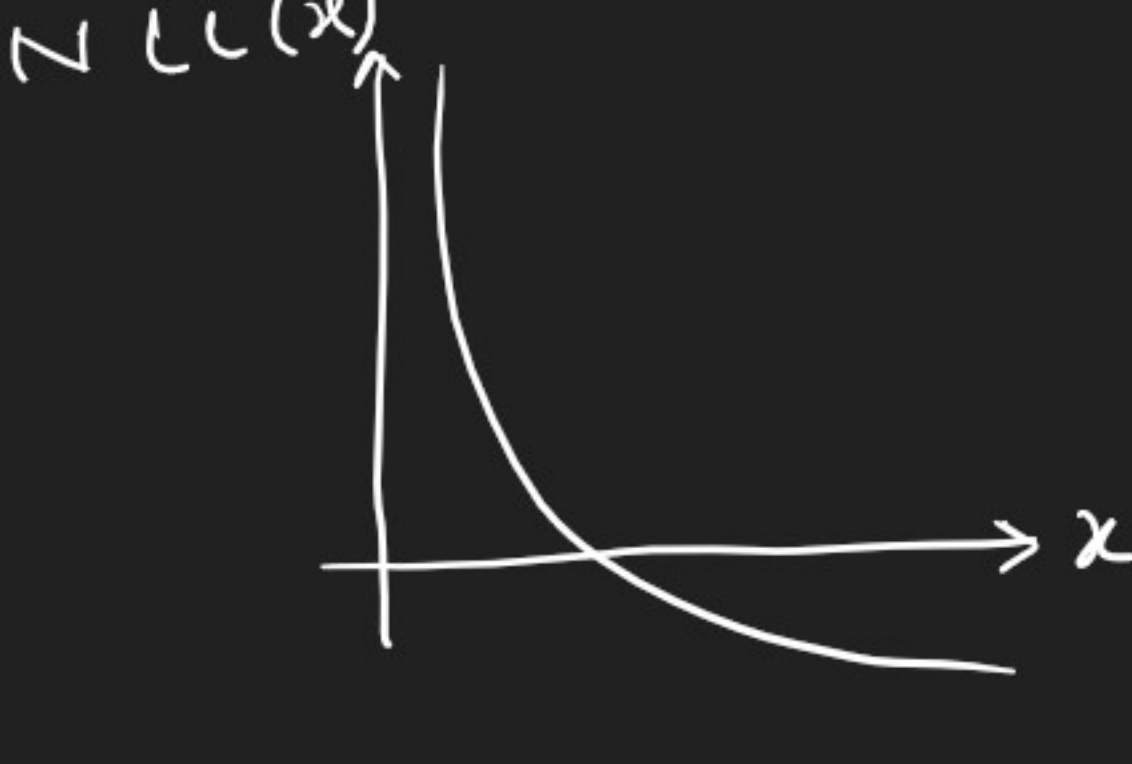
"I had always thought" → GPT model



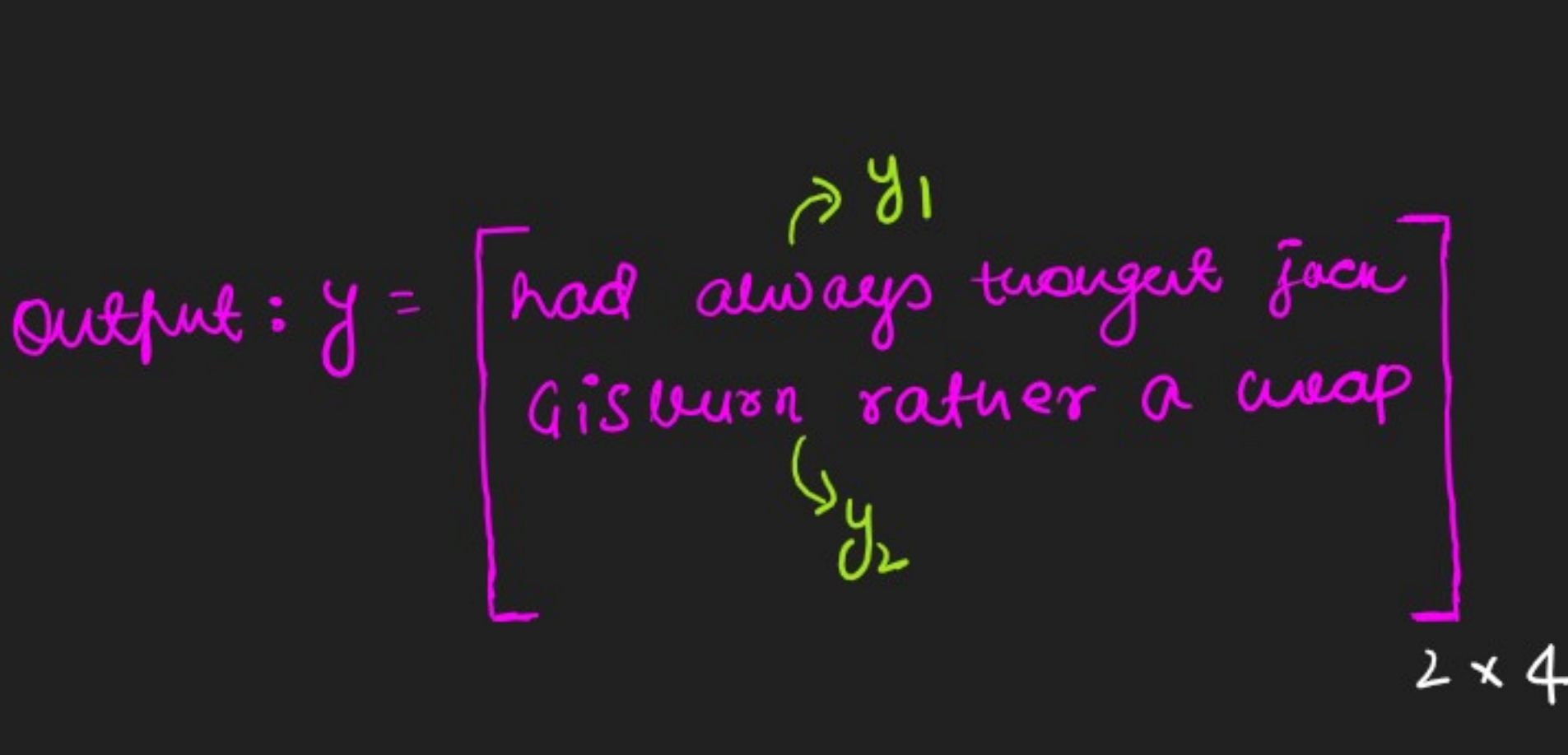
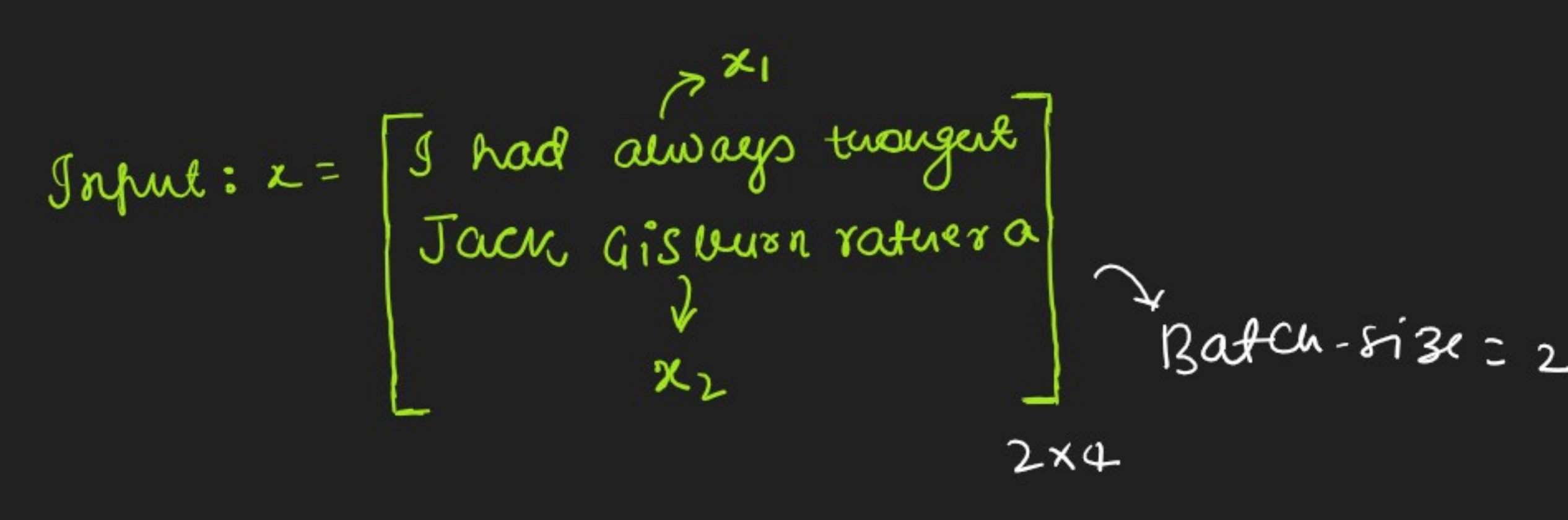
cross entropy loss

$$-\frac{1}{4} \{ \log p_1 + \log p_2 + \log p_3 + \log p_4 \}$$

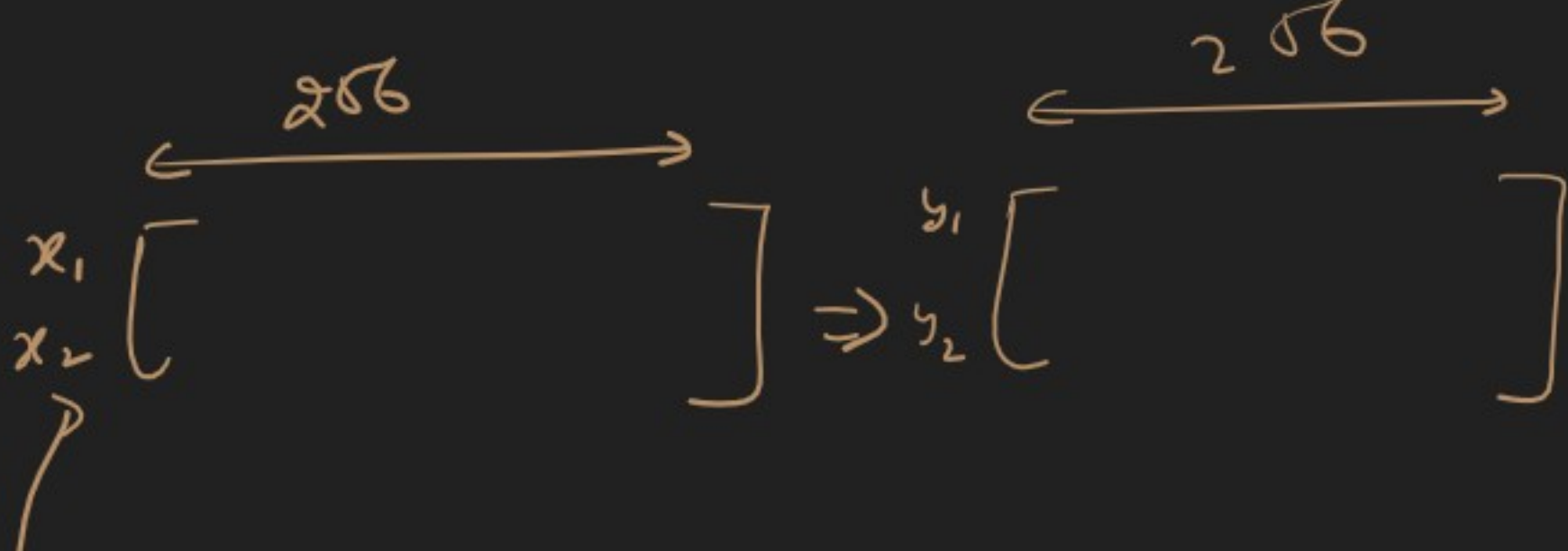
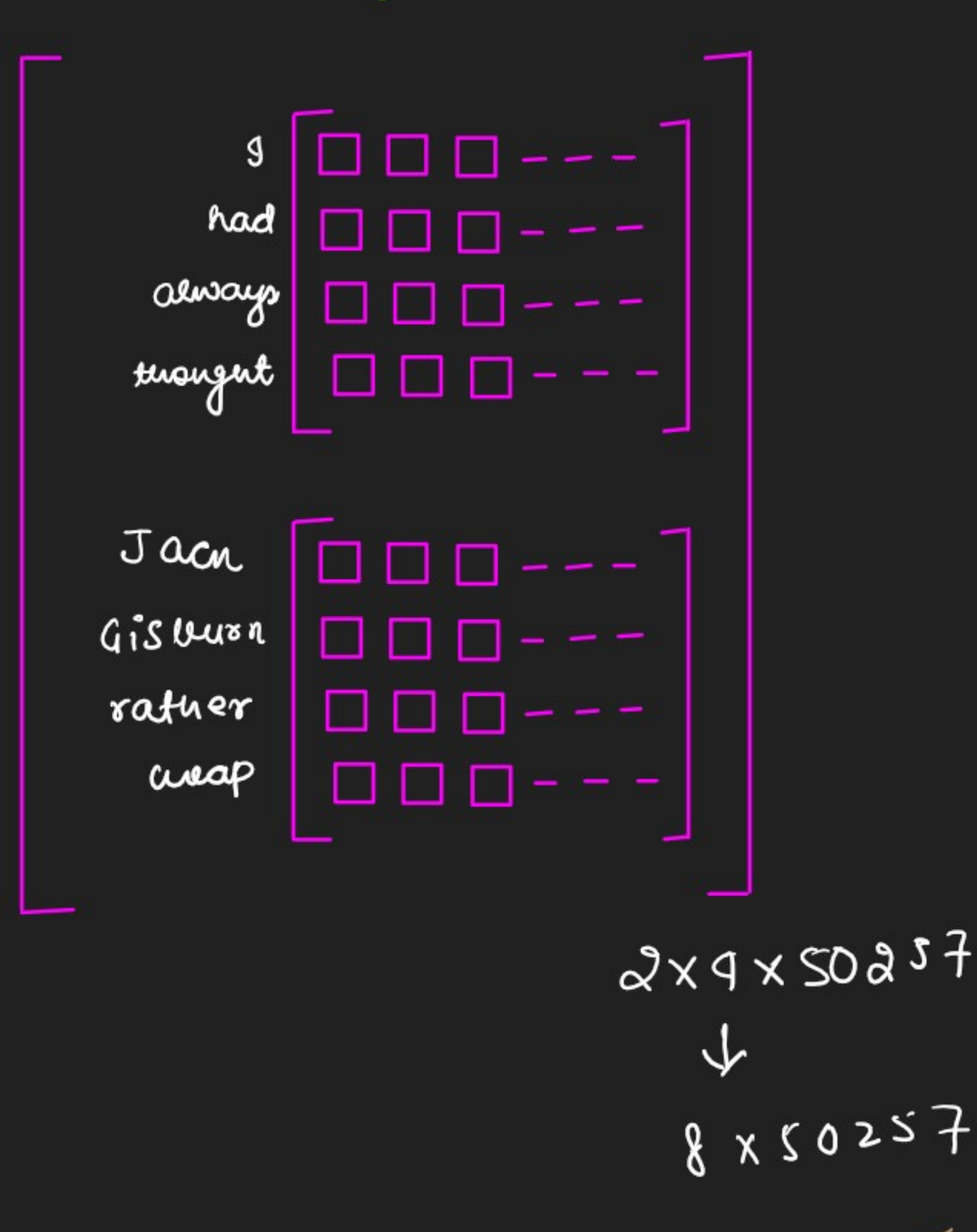
Negative log likelihood



Now, let's us look at how are batch



↓ GPT
logits



```

Train loader:
torch.Size([2, 256]) torch.Size([2, 256])
torch.Size([2, 256]) torch.Size([2, 256])
torch.Size([2, 256]) torch.Size([2, 256])
torch.Size([2, 256]) torch.Size([2, 256])
torch.Size([2, 256]) torch.Size([2, 256])
torch.Size([2, 256]) torch.Size([2, 256])
torch.Size([2, 256]) torch.Size([2, 256])
torch.Size([2, 256]) torch.Size([2, 256])
torch.Size([2, 256]) torch.Size([2, 256])

Validation loader:
torch.Size([2, 256]) torch.Size([2, 256])
  
```

9 training set
batches with
2 sample, 256
token each

One validation
batch with
2 sample 256
token each